VENKATESHWARLU SARANGI, Ph.D

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PROFESSIONAL SUMMARY

Results-driven Data Scientist and AI Engineer with a Ph.D. in Physics and 4+ years of industry experience building and deploying production-grade ML and Generative AI systems. Proven track record translating business requirements into scalable predictive models, end-to-end pipelines, and RAG-based applications using Python, SQL, PyTorch, Scikit-learn, and MLOps tools. Strong expertise in model governance, monitoring, and stakeholder communication, with demonstrated success delivering high-impact solutions in healthcare and materials domains.

TECHNICAL SKILLS

Python, SQL, Scikit-learn, PyTorch, TensorFlow, LLM fine-tuning, RAG systems, vector databases (FAISS, Pinecone, Chroma), MLOps (Docker, MLflow, CI/CD), end-to-end ML pipelines, statistical modeling, feature engineering, model governance, production deployment, stakeholder communication .

WORK EXPERIENCE

DATA SCIENTIST & ML ENGINEER — *Freelancer at XRadio Ltd, HK*

(Aug, 2025 – March, 2026)

Deep Learning and Multimodal Systems for ECG Signal Intelligence

- Built an explainable deep learning model for 12-lead **ECG arrhythmia classification,** addressing challenges from non-stationary signals and inter-patient variability.
- Evaluated multiple **deep learning and vision architectures** including ResNet34, ResNet50, DenseNet, and VGG16, **Vision Transformers (ViT) and Vision-Language Models (VLMs),** multimodal AI systems for healthcare prediction tasks.
- Achieved strong model performance with ResNet34, reaching **AUROC 0.98 and F1-score 0.826** across nine arrhythmia categories.
- Technologies: Python, PyTorch, Scikit-learn, and TensorFlow.

DATA SCIENTIST & RESEARCH ENGINEER — *Gense Technologies Ltd, HK*

(Oct, 2022 – Aug, 2025)

Projects & Technical Experience

- Designed and deployed end-to-end supervised learning pipelines with ML models for clinical prediction tasks, achieving **87% sensitivity** and **>99% specificity** on imbalanced biomedical datasets.
- Engineered feature extraction and signal processing workflows using Python (NumPy, Pandas, Scikit-learn) to improve SNR and diagnostic quality in time-series bio-conductivity data.
- Achieved a **100% improvement** in skin electrode contact impedance without conductive gels, improving signal sensitivity.
- Developed and validated signal enhancement and impedance analysis models for ECG and EIT datasets, improving signal-to-noise ratio and diagnostic reliability.
- Built automated MLOps workflows with Docker, CI/CD, and MLflow for experiment tracking, model versioning, and reproducible deployments across cloud environments. Applied statistical validation frameworks (k-fold cross-validation, ROC-AUC optimization, hypothesis testing) to ensure diagnostic-grade model reliability and regulatory compliance.
- Developed synthetic data augmentation strategies (SMOTE, bootstrapping) to address limited clinical datasets, improving model **generalization by 20%**.
- Proficient in Python, SQL, PyTorch, Scikit-learn, and cloud ML platforms (GCP, Firebase, AWS SageMaker) for scalable production systems.

DATA SCIENTIST AND COMPUTATIONAL ANALYST — *Anvipro IT Solutions, India*

(December, 2021 – June, 2022)

Materials Science & Computational Modeling

- Applied supervised learning and statistical modeling techniques to analyze material properties, identify trends, and support predictive insights.
- Conducted physics-based simulation and data analysis using COMSOL Multiphysics to study material flexibility (>90%), stress distribution, hardness, and structural behavior.
- Performed materials data preprocessing, cleaning, and refinement using MATLAB and Python basic libraries to improve analysis quality and model readiness.
- Improved simulation and analytical workflows through iterative model refinement, results evaluation, and reliability analysis for engineering and materials-focused projects.

- **Technologies:** Python, Data processing, Feature Engineering, EDA,NumPy, Pandas, Scikit-learn, OriginLab, SQL, Power BI, &Statistical Modeling.

PH.D RESEARCH SCHOLAR

CityU HK
(August, 2015 – October 2021)

Materials Engineering / Computational Modeling / Data Analysis

- Architected piezoelectric electrodes **Large electrostrains of 0.2%** and integrated AI-driven computational modeling for material structure analysis using X-ray and neutron diffraction datasets, aligning with predictive AI solutions for operational efficiency.
- Delivered Rietveld refinement and data processing, feature engineering, and regression-based parameter optimization, fine-tuning methods, hyperparameter tuning **Goodness of Fit Chi-squared < 1%** alongside the **Weighted Profile R-factor (R_wp) < 3%**, to extract structural insights from complex datasets, incorporating Machine Learning frameworks like Scikit-learn for enhanced prediction tasks.

JOURNAL PUBLICATIONS

- S. Venkateshwarlu et al., **Nature Communications Physics**, (2020).
- S. Venkateshwarlu et al., **Journal of Materials Research**, Cambridge University Press JMR-0830 (2019).
- S. Nayak, S. Venkateshwarlu, et al., **Journal of the American Ceramic Society**, (2019).
- Frederick, Sarangi Venkateshwarlu, et al., **Chemistry of Materials** (2021).
- A. Pramanick, S. Venkateshwarlu, et al., **Physical Review B**, (2021).
- Liang, Zhuoxin Liu, S. Venkateshwarlu, et al., **Nature**, **Light: Science & Applications**, (2021).
- G. Srinivas, S. Venkateshwarlu, et al., **Applied Physics Letters**, (2015).
- S. Venkateshwarlu, et al., **Journal of Modern Materials**, (2016).
- Pramanick, Venkateshwarlu, **Journal of the European Ceramic Society**, (2023).
- Nirmal and Sarangi Venkateshwarlu et al., **arXiv:2508.10940**, (2025).
- Kwok, W.C., Sarangi Venkateshwarlu et al., **Nature Scientific Reports** (2025).

EDUCATION | TRAINING - CERTIFICATIONS

IIT Bombay India and CityU HK,

- **Ph.D.** Materials Science and Eng (Physics), City University of Hong Kong, (Oct, 2021)
- **M.Tech.** Materials Science and Eng (Physics), IIT Bombay, India, (June, 2013)
- **Generative AI with Large Language Models (LLMs)**
- DeepLearning.AI & Amazon Web Services | Coursera | February 2026
- Skills: NLP, Transformers Architecture (FLAN-T5, BERT, GPT, DeepSeek, Qwen) · RL models · PyTorch · PEFT · Fine-tuning · LoRA · QLoRa · Knowledge Distillation · RAG · LangChain · LangGraph · Deployment · Ollama, vLLM, OpenWebUI, FastAPI, Streamlit , RunPOD, LightningAI, vLLM, HuggingFace, Optuna
- **Machine Learning Specialization** | August 2025
- **Deep Learning and NLP Specialization** | January 2026
- Stanford University & DeepLearning.AI | Coursera
- **End-to-End MLOPS Bootcamp** | Udemy | April 2026